GLOBAL RENEWABLE ENERGY INTELLIGENCE

Data Science Project Report | Notebook review: 23 October 2026

Executive summary

This project examines renewable electricity shares across countries using Our World in Data (OWID) energy data. It combines exploratory analysis, K-Means clustering, and next-year prediction with two machine learning regressors and a persistence baseline. The strongest predictive result on the recorded holdout was the simple baseline, not either machine learning model.

Dataset and preparation

Source: OWID energy-data repository, https://github.com/owid/energy-data . The notebook fetched the CSV dynamically. The saved run contained 23,377 rows, 130 columns and observations through 2025. Filtering to three-letter ISO codes, 2015–2025, and nonmissing renewable electricity share produced 2,191 country-year observations representing 212 distinct country labels. Three-letter codes are a useful filter, but should not be treated as proof that all records are sovereign countries.

Remaining missingness in the cleaned data included wind electricity share (38), hydro electricity share (35), and GDP (871). The notebook did not impute these values globally; clustering and regression used complete-case feature subsets.

Exploratory findings

The unweighted mean renewable electricity share among available country observations increased from 30.80% (2015) to 44.65% (2025), a difference of 13.85 percentage points. This is not the share of total world electricity supplied by renewables. Because country coverage may vary over time, the difference should not be interpreted as a matched-country estimate without further analysis.

Unsupervised learning

K-Means clustering was fitted to the latest available observation per country using standardized solar, wind and hydro electricity shares. The notebook recorded 207 observations, selected five clusters among k=2 to 6 using silhouette score, and obtained a silhouette score of 0.501. Cluster labels are descriptive and not ordinal; the latest available year may differ across countries.

Predictive modeling and holdout evaluation

The regression task predicts next-year renewable electricity share using current renewable, solar, wind, hydro shares and a lagged renewable share. The notebook used 1,439 training records and 279 testing records, with test targets in 2024–2025. A persistence baseline predicted that the next-year share equals

the current-year share.

Method	Holdout MAE (percentage points)
Persistence baseline	1.965
Random Forest	2.144
Histogram Gradient Boosting	2.159

Both machine learning regressors had reported R-squared values of 0.989, but neither beat the persistence baseline on MAE. The R-squared value alone therefore does not justify claiming improved forecast accuracy.

Methodological limitations and next improvements

1. The saved notebook uses the 2024–2025 test period to compare several models; further tuning should use an earlier validation period, keeping a final holdout untouched.

2. Temporal consistency was checked in a separate cell: 1,767 rows had adjacent prior and next years, and the saved check reported zero modeled rows excluded for year gaps. For robust reproducibility, adjacency checks should be integrated directly into model-data construction.

3. The model uses an unweighted cross-country sample and may underrepresent countries with sparse data. Country-specific persistence, year-level coverage, and regional heterogeneity warrant investigation.

4. Cluster fitting selects each country’s latest available year, which can differ. A single common year and sensitivity analysis would strengthen comparability.

5. The results describe historical holdout performance only. No live deployment, future-year forecasts, causal claims, or business impact were established.

Conclusion

The notebook demonstrates data acquisition, cleaning, visualization, unsupervised learning and time-aware predictive benchmarking. Its key finding is that a transparent persistence baseline outperformed the more complex regressors on the recorded test set. Future work should prioritize robust temporal validation, country coverage checks and feature engineering rather than overstating model performance.

Reproducibility: Google Colab notebook Global_Renewable_Energy_Intelligence.ipynb. Data source: https://github.com/owid/energy-data . Values above are taken from outputs saved in the supplied notebook.